

UEI701: DATA ACQUISITION AND SYSTEM DESIGN

L	T	P	Cr
3	0	2	4.0

Course Objectives: To understand concepts of acquiring the data from transducers/input devices, their interfacing and instrumentation system design. To familiarize with different data transfer techniques.

Data Acquisition Techniques and systems: Analog and digital data acquisition, Sensor/Transducer interfacing, Sample and hold circuits, Interference, Grounding and Shielding. Single channel and multichannel data acquisition system, Design of Data Acquisition System, PC-Based data acquisition system, Role of IoT in Data acquisition system

Data Acquisition with Op-Amps: Operational Amplifiers, CMRR, Slew Rate, Gain, Bandwidth. Op amp with negative feedback, Frequency response of Op Amp

Op Amp Circuits: Zero crossing detector, Peak detector, Window detector. Difference Amplifier, Instrumentation Amplifier AD 620, Interfacing of IA with sensors and transducer, Filters in instrumentation circuits.

Data Transfer Techniques: Serial data transmission methods and standards RS 232-C: specifications connection and timing, 4-20 mA current loop, GPIB/IEEE-488, LAN, Universal serial bus, HART protocol, Foundation-Fieldbus, ModBus, Zigbee and Bluetooth. Graphical Interface (GUI) Software for DAS, RTU.

Laboratory Work: Op-amp as a comparator and its application, Integrator and differentiator, Active filters, Simulation of the above applications using different software, Instrumentation Amplifier/AD 620, Interfacing of sensors and transducers using DAQ cards.

Course Learning Outcomes (CLO): After the completion of the course, student will be able to:

1. Compare the elements of data acquisition techniques and differentiate in between single channel and multi-channel
2. Design and simulate signal conditioning circuits
3. Analyze various data transfer techniques used in data acquisition systems
4. Design the data acquisition systems

Text Books:

1. Coughlin, R.F., *Operational Amplifiers and Linear Integrated Circuits*, Pearson Education (2006).
2. Kalsi, H.S., *Electronic Instrumentation*, Tata McGraw Hill (2002).
3. Gayakwad, R.A., *Op-Amp and Linear Integrated Circuits*, Pearson Education (2002).
4. Mathivanan, N., *Microprocessor PC Hardware and Interfacing*, Prentice Hall of India Private Limited (2007).

Reference Books:

1. *Ananad, M.M.S., Electronic Instruments and Instrumentation Technology, Prentice Hall of India Private Limited (2004).*
2. *Murthy, D.V.S., Transducers and Instrumentation, Prentice Hall of India Private Limited (2006).*

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional	30